detector 2, we have no idea which path its signal photon partner took, since there is an equal chance that it was emitted by down-converter L and followed path B–E, or that it was emitted by down-converter R and followed path C–E. Similarly, if an idler is detected by detector 3, it could have been emitted by down-converter L and have traveled path B–F, or by down-converter R and traveled path C–F. Thus, for signal photons whose idlers are detected *by detector 1 or 4, we have which-path information, but for those whose idlers are detected by detector 2 or 3, the which-path information is erased.*

Does this erasure of some of the which-path information—even though we've done *nothing directly* to the signal photons—mean interference effects are recovered? Indeed it does—but only for those signal photons whose idlers wind up in either detector 2 or detector 3. Namely, the totality of impact positions of the signal photons on the screen will look like the data in Figure 7.5a, *showing not the slightest hint of an interference pattern,* as is characteristic of photons that have traveled one path *or* the other. But if we focus on a *subset* of the data points—for example, those signal photons whose idlers entered detector 2—then that subset of points *will* fill out an interference pattern! These signal photons—whose idlers happened, by chance, not to provide any which-path information— act as though they've traveled both paths! If we were to hook up the equipment so that the screen displays a red dot for the position of each signal photon whose idler was detected by detector 2, and a green dot for all others, someone who is color-blind would see no interference pattern, but everyone else would see that the red dots were arranged with bright and dark bands—an interference pattern. The same holds true with detector 3 in place of detector 2. But there would be no such interference pattern if we single out signal photons whose idlers wind up in detector 1 or detector 4, since these are the idlers that yield which-path information about their partners.

These results—which have been confirmed by experiment[5]—are dazzling: by including down-converters that have the potential to provide which-path information, we lose the interference pattern, as in Figure 7.5a. And without interference, we would naturally conclude that each photon went along either the left path or the right path. But we now learn that this would be a hasty conclusion. By carefully eliminating the potential which-path information carried by some of the idlers, we can